

THE UNIVERSITY OF ILLINOIS AT CHICAGO
Department of Chemical Engineering

TEAM AWARD HOMEWORK SET #7

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2017
Professor: Dr. Michael Caracotsios
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Your name: Kevin Joy
Your name: Luke Arnold
Your name: Emmanuel Ampo
Due Date: Wednesday Nov-22, 2017 in class

Number of points.....100
Your score.....

Award: Gift card to each member of the winning team

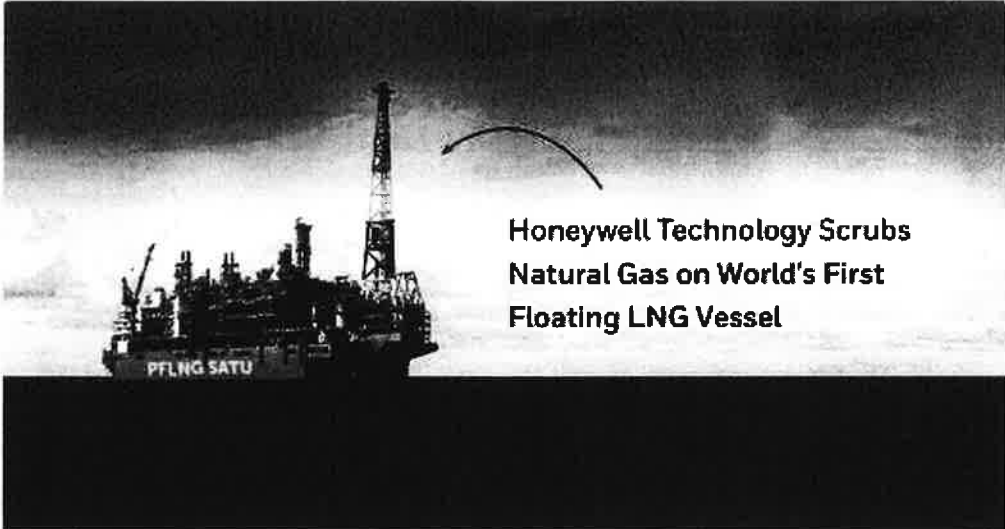
To be considered for the award, the team homework final report must be typed. Homework results should be presented in a logical and concise manner, with all steps in derivation processes [if any] shown very clearly. The ASPEN PFD with critical data must be printed in a full page and included with the homework solution report. Images of the Block Results as well as the Stream Results must also be clearly shown in the homework solution report. You may also want to include anything else (like external references) you believe will increase your chances of winning the award.

Remember: All members of the team that wins a total of three(3) awards will be given **100** points towards the final examination and will be excused from taking it.

Problem 1(100 pts): Aspen Plus Simulation of CO₂ Removal from Syngas/Natural Gas

Honeywell UOP
Honeywell
UOP

PETRONAS' floating liquefied natural gas (FLNG) facility, the first of its kind in the world, uses Honeywell UOP's Amine Guard FS™ technology, specially adapted for marine applications



**Honeywell Technology Scrubs
Natural Gas on World's First
Floating LNG Vessel**

World's first floating LNG facility-PETRONAS

Syngas is usually produced from gasification of coal or steam methane reforming. Depending on the source of production it contains various amounts for carbon dioxide that needs to be removed prior to further purification for production of hydrogen. The removal of CO₂ is usually accomplished with chemical solvents, such as amines. The most common amine solvent is MDEA, methyl di-ethanolamine. A typical process for the removal of CO₂ is shown in the process flowsheet diagram below.

other data, you will need to do some research on your own. Make sure everyone in the team does his/her own part.

OBJECTIVES OF THIS HOMEWORK

1. Prepare an ASPEN Plus Simulation for this process. You may start with the basic template that I have created for you and contains the physical properties and the amine chemistry
2. Your objective is to calculate the optimal amounts of the lean solvent and the reboiler duty in order to make the operating cost reasonable and to obtain a molar composition of carbon dioxide in the clean gas of 1%. Assume that cost of cooling is **0.2\$/GJ**, the cost of heating is **\$5.5/GJ**, and cost of electricity is **\$0.06/kWh**.
3. The lean solvent has a molar water to amine ratio of 6.615. The load is something that you have to decide
4. The team that arrives at the smallest cost and meets the specifications of the clean gas will get the award.

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I INTRODUCTION

Steam methane reforming or the gasification of coal are two of the most common processes that produce Syngas. Syngas, which is an abbreviation for synthesis gas, is a fuel gas mixture consisting primarily of hydrogen, carbon monoxide, and carbon dioxide. This report serves as a proposal for the removal of CO₂ through the production of Syngas. Our analysis of the production process was performed using a specified feed line paired with two primary unit operations, an absorber and a stripper. The objective was to model the removal process based on the specifications provided, while calculating the annual operating expense with respect to electricity, cooling, and heating utilities. Assuming that the ELECCTRL equation of state adequately describes the physical properties of the operation, paired with the method outlined in this report, our team was able to generate a value of \$1,194,205/year based on the performances of the energy intensive units outlined in this report.

II MAIN FLOWSHEET

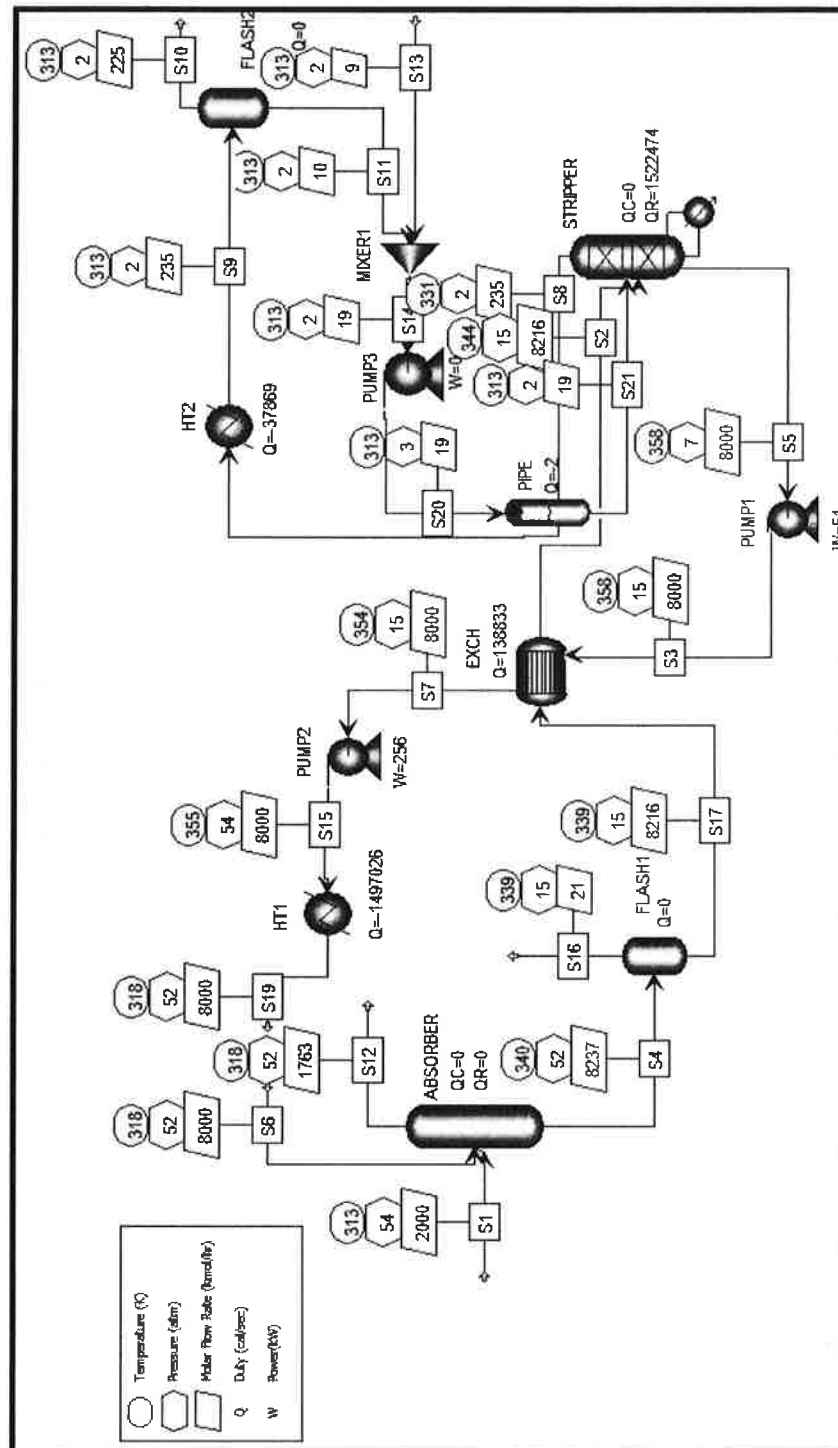


Figure 1: Main Flowsheet

Emmanuel Ampo

Luke Arnold

Kevin Joy

III STREAM INPUT

☒ Mixed ☐ CI Solid ☐ NC Solid **Flash Options** ☐ EO Options ☐ Costing ☐ Information

Specifications

Flash Type: **Temperature** **Pressure**

State variables

Temperature: **40** **C**

Pressure: **55** **bar**

Vapor fraction:

Total flow basis: **Mole**

Total flow rate: **2000** **kmol/hr**

Solvent:

Composition

Mole-Frac

Component	Value
CO	0.291
CO2	0.1
H2	0.598
H2O	0.002
N2	0.003
AR	0.001
CH4	0.005
C5H13NO2	0
MDEA+	0
H3O+	0
OH-	0
HCO3-	0
CO3--	0
Total	1

☒ Reference Temperature
☒ Component Attributes

Figure 2: Stream 1 (S1) Input

Main Flowsheet × **S6 (MATERIAL)** × S13 (MATERIAL) × +

☒ Mixed ☐ CI Solid ☐ NC Solid **Flash Options** ☐ EO Options ☐ Costing ☐ Information

Specifications

Flash Type: **Temperature** **Pressure**

State variables

Temperature: **45** **C**

Pressure: **53** **bar**

Vapor fraction:

Total flow basis: **Mole**

Total flow rate: **8000** **kmol/hr**

Solvent:

Composition

Mole-Frac

Component	Value
CO	0
CO2	0.065071
H2	0
H2O	0.812155
N2	0
AR	0
CH4	0
C5H13NO2	0.122775
MDEA+	0
H3O+	0
OH-	0
HCO3-	0
CO3--	0
Total	1

Figure 3: Stream 6 (S6) Input

Main Flowsheet × S6 (MATERIAL) × S13 (MATERIAL) × +

☒ Mixed
 ☐ CI Solid
 ☐ NC Solid
 ☐ Flash Options
 ☐ EO Options
 ☐ Costing
 ☐ Information

Specifications

Flash Type Temperature Pressure Composition

State variables

Temperature:	39.95	C
Pressure:	2	bar
Vapor fraction:		
Total flow basis:	Mole	
Total flow rate:	9.136	kmol/hr
Solvent:		

Mole-Flow kmol/hr

Component	Value
CO	
CO2	
H2	
H2O	3600
N2	
AR	
CH4	
C5H13NO2	
MDEA+	
H3O+	
OH-	
HCO3-	
CO3--	
Total	3600

Figure 4: Stream 13 (S13) Input

IV EQUIPMENT INPUT

RADFRACs

ABSORBER

☒ Configuration	☒ Streams	☒ Pressure	Condenser	Reboiler	3-Phase	Information
Setup options						
Calculation type:		Equilibrium				
Number of stages:		15		Stage Wizard		
Condenser:		None				
Reboiler:		None				
Valid phases:		Vapor-Liquid				
Convergence:		Standard				
Operating specifications						
Free water reflux ratio:		0		Feed Basis		

Figure 5: ABSORBER Configuration Tab Input

☒ Configuration	☒ Streams	☒ Pressure	Condenser	Reboiler	3-Phase	Information				
Feed streams										
	Name	Stage	Convention							
▶	S6	1	On-Stage							
	S1	15	On-Stage							
Product streams										
	Name	Stage	Phase	Basis	Flow	Units	Flow Ratio	Feed Specs		
	S4	15	Liquid	Mole		kmol/hr		Feed basis		
	S12	1	Vapor	Mole		kmol/hr		Feed basis		
Pseudo streams										
	Name	Pseudo Stream Type	Stage	Internal Phase	Reboiler Phase	Reboiler Conditions	Pumparound ID	Pumparound Conditions	Flow	Units

Figure 6: ABSORBER Streams Tab Input

☒ Configuration
 ☒ Streams
 ☒ Pressure
 Condenser Reboiler 3-Phase Information

View: **Top / Bottom**

Top stage / Condenser pressure

Stage 1 / Condenser pressure: **53** bar

Stage 2 pressure (optional)

☒ Stage 2 pressure: bar
☐ Condenser pressure drop: bar

Pressure drop for rest of column (optional)

☒ Stage pressure drop: bar
☐ Column pressure drop: bar

Figure 7: ABSORBER Pressure Tab Input

STRIPPER

☒ Configuration
 ☒ Streams
 ☒ Pressure
 Condenser ☒ Reboiler 3-Phase Information

Setup options

Calculation type: **Equilibrium**

Number of stages: **18** Stage Wizard

Condenser: **None**

Reboiler: **Kettle**

Valid phases: **Vapor-Liquid**

Convergence: **Standard**

Operating specifications

Reboiler duty: **2.175e+07** Btu/hr

Free water reflux ratio: **0** Feed Basis

Figure 8: STRIPPER Configuration Tab Input

☒ Configuration	☒ Streams	☒ Pressure	Condenser	☒ Reboiler	3-Phase	Information			
Feed streams									
Name	Stage	Convention							
S2	1	On-Stage							
S21	1	Above-Stage							
Product streams									
Name	Stage	Phase	Basis	Flow	Units	Flow Ratio	Feed Specs		
S8	1	Vapor	Mole		kmol/hr		Feed basis		
S5	18	Liquid	Mole		kmol/hr		Feed basis		
Pseudo streams									
Name	Pseudo Stream Type	Stage	Internal Phase	Reboiler Phase	Reboiler Conditions	Pumparound ID	Pumparound Conditions	Flow	Units

Figure 9: STRIPPER Streams Tab Input

☒ Configuration	☒ Streams	☒ Pressure	Condenser	☒ Reboiler	3-Phase	Information
View: Top / Bottom						
Top stage / Condenser pressure						
Stage 1 / Condenser pressure:		2	bar			
Stage 2 pressure (optional)						
☒ Stage 2 pressure:			bar			
☐ Condenser pressure drop:			bar			
Pressure drop for rest of column (optional)						
☒ Stage pressure drop:		0.3	bar			
☐ Column pressure drop:			bar			

Figure 10: STRIPPER Pressure Tab Input

FLASH2s

FLASH1

✓ Specifications		Flash Options	Entrainment	PSD	Utility	Information
Flash specifications						
Flash Type:	Pressure		Duty			
Temperature:			C			
Pressure:	15		bar			
Duty:	0		Watt			
Vapor fraction:						
Valid phases						
Vapor-Liquid						

Figure 11: FLASH1 Specifications Tab Input

FLASH2

✓ Specifications		Flash Options	Entrainment	PSD	Utility	Information
Flash specifications						
Flash Type:	Pressure		Duty			
Temperature:	75		C			
Pressure:	2		bar			
Duty:	0		Watt			
Vapor fraction:						
Valid phases						
Vapor-Liquid						

Figure 12: FLASH2 Specifications Tab Input

HEATERS*HT1*

Specifications		Flash Options	Utility	Information
Flash specifications				
Flash Type:	Temperature Pressure			
Temperature:	45	C		
Temperature change:		C		
Degrees of superheating:		C		
Degrees of subcooling:		C		
Pressure:	53	bar		
Duty:		Watt		
Vapor fraction:				
Pressure drop correlation parameter:				
Valid phases				
Vapor-Liquid				

Figure 13: HT1 Specifications Tab Input

HT2

Specifications		Flash Options	Utility	Information
Flash specifications				
Flash Type:	Temperature Pressure			
Temperature:	40	C		
Temperature change:		C		
Degrees of superheating:		C		
Degrees of subcooling:		C		
Pressure:	2	bar		
Duty:		Watt		
Vapor fraction:				
Pressure drop correlation parameter:				
Valid phases				
Vapor-Liquid				

Figure 14: HT2 Specifications Tab Input

PUMPS

PUMP1

☒ Specifications	Calculation Options	Flash Options	Utility	Information
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Model

☒ Pump ☐ Turbine

Pump outlet specification

☒ Discharge pressure: 15 bar

☐ Pressure increase: bar

☐ Pressure ratio:

☐ Power required: Watt

☐ Use performance curve to determine discharge conditions 

Efficiencies

Pump: 1 Driver:

Figure 15: PUMP1 Specifications Tab Input

PUMP2

☒ Specifications	Calculation Options	Flash Options	Utility	Information
---	---------------------	---------------	---------	-------------

Model

☒ Pump ☐ Turbine

Pump outlet specification

☒ Discharge pressure: 55 bar

☐ Pressure increase: bar

☐ Pressure ratio:

☐ Power required: Watt

☐ Use performance curve to determine discharge conditions 

Efficiencies

Pump: 1 Driver:

Figure 16: PUMP2 Specifications Tab Input

PUMP3

☒ Specifications	Calculation Options	Flash Options	Utility	Information
---	---------------------	---------------	---------	-------------

Model

☒ Pump ☐ Turbine

Pump outlet specification

☒ Discharge pressure: **3.2** **bar**

☐ Pressure increase: **bar**

☐ Pressure ratio:

☐ Power required: **Watt**

☐ Use performance curve to determine discharge conditions 

Efficiencies

Pump: **1** Driver:

Figure 17: PUMP3 Specifications Tab Input

HEAT EXCHANGERS

EXCH

☒ Specifications	Streams	LMTD	Pressure Drop	☒ U Methods	Film Coeff
---	---------	------	---------------	--	------------

Calculation

☒ Shortcut

☐ Detailed

☐ Rigorous

Size Shell&Tube

☐ Use Design Template File

Rigorous Model

☐ Shell&Tube

☐ AirCooled

☐ Plate

Transfer UA to shortcut

Flow arrangement

Hot fluid:

Flow direction: **Countercurrent**

No. shells in series: **1**

☐ Calculate number of shells

Type: **Design**

Exchanger type: **Heat exchanger**

Exchanger specification

Specification: **Hot / cold outlet temperature approach**

Value: **10** **C**

Exchanger area: **sqm**

Constant UA: **J/sec-K**

Minimum temperature approach: **1** **C**

Figure 18: EXCH Specifications Tab Input

PIPEsPIPE

☒ Pipe Parameters	Thermal Specification	Fittings1	Fittings2	Flash Options	Solids Conveying	Information
--	-----------------------	-----------	-----------	---------------	------------------	-------------

☒ Fluid flow
☐ Solids conveying

Length
 Pipe length: **38** **ft**

Diameter
☐ Inner diameter: 6.065 in
☒ Use pipe schedules
☐ Compute using user subroutine

Pipe schedules
 Material: **CARBON-STEEL**
 Schedule: **40**
 Nom diameter: **6-IN**

Elevation
☒ Pipe rise: **38** **ft**
☐ Pipe angle: **rad**

Options
 Roughness: **4.572e-05** **meter**
 Erosional velocity coefficient: **100**

Figure 19: PIPE Pipe Parameters Tab Input

MIXERsMIXER1

☒ Flash Options	Information
--	-------------

Mixer specifications
 Pressure: **2** **bar**
 Valid phases: **Vapor-Liquid**

Temperature estimate
39.95 **C**

Convergence parameters
 Maximum iterations: **30**
 Error tolerance: **0.0001**

Figure 20: MIXER1 Flash Options Tab Input

COMPONENTS

Component ID	Type	Component name	Alias
CO	Conventional	CARBON-MONOXIDE	CO
CO2	Conventional	CARBON-DIOXIDE	CO2
H2	Conventional	HYDROGEN	H2
H2O	Conventional	WATER	H2O
N2	Conventional	NITROGEN	N2
AR	Conventional	ARGON	AR
CH4	Conventional	METHANE	CH4
C5H13NO2	Conventional	METHYL-DIETHANOLAMINE	C5H13NO2
MDEA+	Conventional	MDEA+	C5H14NO2+
H3O+	Conventional	H3O+	H3O+
OH-	Conventional	OH-	OH-
HCO3-	Conventional	HCO3-	HCO3-
CO3--	Conventional	CO3--	CO3-2

Figure 21: Components used for the Simulation

VI PHYSICAL PROPERTIES

☒ Global Flowsheet Sections Referenced Information

Property methods & options
 Method filter: **COMMON**
 Base method: **ELECNRTL**
 Henry components: **GLOBAL**

Petroleum calculation options
 Free-water method: **STEAM-TA**
 Water solubility: **3**

Electrolyte calculation options
 Chemistry ID: **GLOBAL**
☐ Use true components

Method name: **ELECNRTL** **Methods Assistant...**

☐ Modify

Vapor EOS: **ESRK**
 Data set: **1**
 Liquid gamma: **GMENRTL**
 Data set: **1**
 Liquid molar enthalpy: **HLMXELC**
 Liquid molar volume: **VLMXELC**

☐ Heat of mixing
☒ Poynting correction
☐ Use liquid reference state enthalpy

Figure 22: Property Methods

- Parameters
 - Pure Components
 - Binary Interaction
 - ANDKJ-1
 - ANDMJ-1
 - HENRY-1
 - HOCETA-1
 - MLQKJ-1
 - MUKJ-1
 - MULJ-1
 - NRTL-1
 - RKTKJ-1
 - VLCLK-1
 - VLQKJ-1
 - Electrolyte Pair
 - GMELCC-1
 - GMELCD-1
 - GMELCE-1
 - GMELCN-1

Figure 23: Binary Interaction and Electrolyte Pair Parameters

VII BLOCK RESULTSRADFRACsABSORBER

Condenser/Top Stage Performance			Reboiler/Bottom Stage Performance		
Name	Value	Units	Name	Value	Units
Temperature	45.0528007	C	Temperature	66.7991306	C
Heat duty	0	GJ/hr	Heat duty	0	GJ/hr
Distillate rate	1763.12034	kmol/hr	Bottoms rate	8236.87966	kmol/hr
Reflux rate	8084.49203	kmol/hr	Boilup rate	1983.12084	kmol/hr
Reflux ratio	4.58533195		Boilup ratio	0.240761177	

Table 1: ABSORBER Results Summary Tab

Summary	Balance	Split Fraction	Reboiler	Utilities	Stage Utilities	Status
Total	Units	In	Out	Rel. diff		
Mole-flow	kmol/hr	10000	10000	0		
Mass-flow	kg/hr	284998	284998	4.36868e-13		
Enthalpy	Watt	-7.4459e+08	-7.44602e+08	1.59071e-05		

Figure 24: ABSORBER Results Balance Tab

Summary	Balance	Split Fraction	Reboiler
Component	S12	S4	
CO	0.927633	0.0723674	
CO2	0.0247837	0.975216	
H2	0.998673	0.00132652	
H2O	0.000505482	0.999495	
N2	0.999172	0.000827634	
AR	0.821378	0.178622	
CH4	0.00421149	0.995789	
C5H13NO2	1.6387e-07	1	

Figure 25: ABSORBER Results Split Fraction Tab

STRIPPER

Condenser/Top Stage Performance			Reboiler/Bottom Stage Performance		
Name	Value	Units	Name	Value	Units
Temperature	57.6688352	C	Temperature	85.094436	C
Heat duty	0	GJ/hr	Heat duty	22.94746481	GJ/hr
Distillate rate	234.607336	kmol/hr	Bottoms rate	8000.05522	kmol/hr
Reflux rate	8111.56216	kmol/hr	Boilup rate	204.727389	kmol/hr
Reflux ratio	34.5750576		Boilup ratio	0.025590747	

Table 2: STRIPPER Results Summary Tab

Summary	Balance	Split Fraction	Reboiler	Utilities	Stage Utilities	✓ Status
	Total	Units	In	Out	Rel. diff	
► Mole-flow		kmol/hr	8234.66	8234.66	0	
Mass-flow		kg/hr	266098	266098	-8.53107e-15	
Enthalpy		Watt	-7.25989e+08	-7.19615e+08	-0.00878015	

Figure 26: STRIPPER Results Balance Tab

Summary	Balance	Split Fraction	Reboiler
Component	S8	S5	
► CO	1	0	
CO2	0.254814	0.745186	
H2	1	0	
H2O	0.00278133	0.997219	
N2	1	0	
AR	1	2.09956e-14	
CH4	0.997168	0.00283189	
C5H13NO2	8.88255e-07	1	

Figure 27: STRIPPER Results Split Fraction Tab

FLASH2sFLASH1

Summary	Balance	Phase Equilibrium	Utility Usage	Status
Outlet temperature:	66.3193	C		
Outlet pressure:	15	bar		
Vapor fraction:	0.00255261			
Heat duty:	0	GJ/hr		
Net duty:	0	GJ/hr		
1st liquid / Total liquid:	1			

Figure 28: FLASH1 Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	8236.88	8236.88	2.21E-16
Mass-flow	kg/hr	266388	266388	2.19E-16
Enthalpy	GJ/hr	-2614.01	-2614.01	1.63E-10

Table 3: FLASH1 Results Balance Tab

Component	F	X	Y	K
CO	0.005113322	0.003519406	0.62794456	178.423298
CO2	0.08531254	0.084862474	0.261178323	3.07766094
H2	0.000192611	6.81E-06	0.072793689	10682.3841
H2O	0.788884568	0.790862132	0.016140624	0.020408874
N2	6.03E-07	1.38E-08	0.000230787	16725.196
AR	4.34E-05	3.65E-05	0.002713853	74.2764268
CH4	0.001208939	0.001163417	0.018997078	16.3286845
C5H13NO2	0.119244046	0.119549206	1.09E-06	9.08E-06

Table 4: FLASH1 Results Phase Equilibrium Tab

FLASH2

Summary	Balance	Phase Equilibrium	Utility Usage	Status
Outlet temperature:	40	C		
Outlet pressure:	2	bar		
Vapor fraction:	0.958772			
Heat duty:	0	GJ/hr		
Net duty:	0	GJ/hr		
1st liquid / Total liquid:	1			

Figure 29: FLASH2 Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	234.607336	234.607336	2.42E-16
Mass-flow	kg/hr	9120.77586	9120.77586	3.99E-16
Enthalpy	GJ/hr	-78.56270352	-78.56270352	-8.84E-12

Table 5: FLASH2 Results Balance Tab

Component	F	X	Y	K
CO	0.123263	0.000358	0.128548	359.2152
CO2	0.757278	0.000751	0.789809	1051.652
H2	0.000239	6.60E-09	0.000249	37737.57
H2O	0.077253	0.997431	0.037685	0.037782
N2	4.83E-07	9.84E-12	5.04E-07	51201.17
AR	0.00128	8.73E-06	0.001335	152.7825
CH4	0.040683	0.001361	0.042374	31.13846
C5H13NO2	3.72E-06	9.02E-05	2.80E-12	3.11E-08

Table 6: FLASH2 Results Phase Equilibrium Tab

HEATERSHT1

Summary	Balance	Phase Equilibrium	Utility Usage	Status
Outlet temperature:	45	C		
Outlet pressure:	53	bar		
Vapor fraction:	0			
Heat duty:	-22.5639	GJ/hr		
Net duty:	-22.5639	GJ/hr		
1st liquid / Total liquid:	1			
Pressure-drop correlation parameter:				

Figure 30: HT1 Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	8000.051	8000.051	0
Mass-flow	kg/hr	256976.8	256976.8	0
Enthalpy	GJ/hr	-2513.61	-2536.17	0.008897

Table 7: HT1 Results Balance Tab

Component	F	X	Y	K
CO2	0.064945	0.064945	0.746868	0.165391
H2O	0.812278	0.812278	0.252456	0.004439
CH4	3.39E-06	3.39E-06	0.000564	2.401876
C5H13NO2	0.122774	0.122774	0.000112	1.27E-05

Table 8: HT1 Results Phase Equilibrium Tab

HT2

Summary	Balance	Phase Equilibrium	Utility Usage	Status
Outlet temperature:	40	C		
Outlet pressure:	2	bar		
Vapor fraction:	0.958772			
Heat duty:	-0.570777	GJ/hr		
Net duty:	-0.570777	GJ/hr		
1st liquid / Total liquid:	1			
Pressure-drop correlation parameter:				

Figure 31: HT2 Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	234.6073	234.6073	1.21E-16
Mass-flow	kg/hr	9120.776	9120.776	0
Enthalpy	GJ/hr	-77.9919	-78.5627	0.007265

Table 9: HT2 Results Balance Tab

Component	F	X	Y	K
CO	0.123263	0.000358	0.128548	359.2152
CO2	0.757278	0.000751	0.789809	1051.653
H2	0.000239	6.60E-09	0.000249	37737.57
H2O	0.077253	0.997431	0.037685	0.037782
N2	4.83E-07	9.84E-12	5.04E-07	51201.17
AR	0.00128	8.73E-06	0.001335	152.7825
CH4	0.040683	0.001361	0.042374	31.13846
C5H13NO2	3.72E-06	9.02E-05	2.80E-12	3.11E-08

Table 10: HT2 Results Phase Equilibrium Tab

PUMPSPUMP1

Summary	Balance	Performance Curve	Utility Usage
Fluid power:	50.8709068	kW	
Brake power:	50.8709	kW	
Electricity:	50.8709	kW	
Volumetric flow rate:	0.0643936	cum/sec	
Pressure change:	7.9	bar	
NPSH available:	0	J/kg	
NPSH required:			
Head developed:	712.653	J/kg	
Pump efficiency used:	1		
Net work required:	50.8709	kW	
Outlet pressure:	15	bar	

Figure 32: PUMP1 Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	8000.055	8000.055	0
Mass-flow	kg/hr	256976.9	256976.9	0
Enthalpy	kW	-697951	-697900	-7.29E-05

Table 11: PUMP1 Results Balance Tab

PUMP2

Summary	Balance	Performance Curve	Utility Usage
Fluid power:	256.411991	kW	
Brake power:	256.412	kW	
Electricity:	256.412	kW	
Volumetric flow rate:	0.064103	cum/sec	
Pressure change:	40	bar	
NPSH available:	873.002	J/kg	
NPSH required:			
Head developed:	3592.09	J/kg	
Pump efficiency used:	1		
Net work required:	256.412	kW	
Outlet pressure:	55	bar	

Figure 33: PUMP2 Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	8000.051	8000.051	0
Mass-flow	kg/hr	256976.8	256976.8	0
Enthalpy	kW	-698481	-698224	-0.00037

Table 12: PUMP2 Results Balance Tab

PUMP3

Summary	Balance	Performance Curve	Utility Usage	✓
Fluid power:	0.0114438017	kW		
Brake power:	0.0114438	kW		
Electricity:	0.0114438	kW		
Volumetric flow rate:	9.5365e-05	cum/sec		
Pressure change:	1.2	bar		
NPSH available:	95.2256	J/kg		
NPSH required:				
Head developed:	121.482	J/kg		
Pump efficiency used:	1			
Net work required:	0.0114438	kW		
Outlet pressure:	3.2	bar		

Figure 34: PUMP3 Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	18.80845	18.80845	0
Mass-flow	kg/hr	339.1271	339.1271	0
Enthalpy	kW	-1487.72	-1487.71	-7.69E-06

Table 13: PUMP3 Results Balance Tab

HEAT EXCHANGERS*EXCH*

Summary	Balance	Exchanger Details	Pres Drop/Velocities	Zones	Utility Usage	Status
Heatx results						
Calculation Model	Shortcut					
	Inlet			Outlet		
Hot stream:	S3		S7			
Temperature:	85.3407	C	81.0094	C		
Pressure:	15	bar	15	bar		
Vapor fraction:	0		0			
1st liquid / Total liquid	1		1			
Cold stream:	S17		S2			
Temperature:	66.3193	C	71.0154	C		
Pressure:	15	bar	15	bar		
Vapor fraction:	0		0.000881775			
1st liquid / Total liquid	1		1			
Heat duty:	2.09256	GJ/hr				

Figure 35: EXCH Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	16215.91	16215.9	2.78E-07
Mass-flow	kg/hr	522735.4	522735.3	1.58E-07
Enthalpy	GJ/hr	-5122.74	-5122.74	-2.47E-07

Table 14: EXCH Results Balance Tab

Calculated heat duty:	2.092556	GJ/hr
Required exchanger area:	47.13912	sqm
Actual exchanger area:	47.13912	sqm
Percent over (under) design:	0	
Average U (Dirty):	850	Watt/sqm-K
Average U (Clean):		
UA:	40068.26	J/sec-K
LMTD (Corrected):	14.50688	C
LMTD correction factor:	1	
Thermal effectiveness:		
Number of transfer units:		
Number of shells in series:	1	
Number of shells in parallel:		

Table 15: EXCH Results Exchanger Details Tab

Summary	Balance	Exchanger Details	Pres Drop/Velocities	Zones	Utility Usage	Status
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View: **Zone summary**

Profiles

Zone	Hot-side Temp	Cold-side Temp	LMTD	U	Duty	Area	UA
	C	C	C	Watt/sqm	Watt	sqm	J/sec-K
1	81.0094	71.0154	14.5069	850	581266	47.1391	40068.3

Figure 36: EXCH Results Zones Tab

PIPEsPIPE

Summary	Streams	Balance	Profiles	Properties	Property Grid
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Total pressure drop:	1.12202403	bar
Frictional pressure drop:	4.77023e-07	bar
Elevation pressure drop:	1.12202	bar
Acceleration:	0	bar
Heat duty:	-3.35943e-05	GJ/hr
Equivalent length:	11.5824	meter

Figure 37: PIPE Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	18.80845	18.80846	-4.49E-07
Mass-flow	kg/hr	339.1271	339.1272	-4.52E-07
Enthalpy	GJ/hr	-5.35576	-5.3558	6.72E-06

Table 16: PIPE Results Balance Tab

MIXERsMIXER1

Summary	Balance	☒ Status
Outlet temperature:	39.9756185	C ▼
Outlet pressure:	2	bar ▼
Vapor fraction:	0	
1st liquid/Total liquid:	1	

Figure 38: MIXER1 Results Summary Tab

Total	Units	In	Out	Rel. diff
Mole-flow	kmol/hr	18.80845	18.80845	1.89E-16
Mass-flow	kg/hr	339.1271	339.1271	1.68E-16
Enthalpy	kW	-1487.72	-1487.72	-4.36E-13

Table 17: MIXER1 Results Balance Tab

VIII STREAM RESULTS

From	Units	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
To												
Substream: MIXED		ABSORBER	STRIPPER	EXCH	PUMP1	FLASH1	ABSORBER	PUMP1	STRIPPER	HT2	FLASH2	MIXER1
Phase:		Mixed	Mixed	Liquid	Liquid	Liquid	Liquid	Liquid	Vapor	Mixed	Vapor	Liquid
Component Mole Flow												
CO	KMOL/HR	582	28.91493	1.03E-19	42.11782	1.03E-19	0	1.03E-19	28.91839	28.91839	28.91493	0.003461
CO2	KMOL/HR	200	697.2177	519.5624	702.7091	519.5624	520.5675	519.5623	177.6626	177.6626	177.6553	0.007264
H2	KMOL/HR	1196	0.055986	2.98E-54	1.586514	2.98E-54	0	2.98E-54	0.055986	0.055986	0.055986	6.38E-08
H2O	KMOL/HR	4	6497.608	6498.267	6497.947	6498.267	6497.234	6498.264	18.12418	18.12418	8.476597	9.647587
N2	KMOL/HR	6	0.000113	2.29E-60	0.004956	2.29E-60	0	2.29E-60	0.000113	0.000113	0.000113	9.52E-11
AR	KMOL/HR	2	0.300184	6.31E-15	0.357244	6.31E-15	0	6.31E-15	0.300269	0.300269	0.300184	8.45E-05
CH4	KMOL/HR	10	9.558461	0.027108	9.957885	0.027108	0	0.027108	9.544516	9.544516	9.531353	0.013162
C5H13NO2	KMOL/HR	0	982.1988	982.1988	982.1989	982.1988	982.199	982.1988	0.000872	0.000872	6.31E-10	0.000872
MDEA+	KMOL/HR	0	0	0	0	0	0	0	0	0	0	0
H3O+	KMOL/HR	0	0	0	0	0	0	0	0	0	0	0
OH-	KMOL/HR	0	0	0	0	0	0	0	0	0	0	0
HCO3-	KMOL/HR	0	0	0	0	0	0	0	0	0	0	0
CO3--	KMOL/HR	0	0	0	0	0	0	0	0	0	0	0
Component Mole Fraction												
CO		0.291	0.003519	1.28E-23	0.005113	1.28E-23	0	1.28E-23	0.123263	0.123263	0.128548	0.000358
CO2		0.1	0.084862	0.064945	0.085313	0.064945	0.065071	0.064945	0.757278	0.757278	0.789809	0.000751
H2		0.598	6.81E-06	3.73E-58	0.000193	3.73E-58	0	3.73E-58	0.000239	0.000239	0.000249	6.6E-09
H2O		0.002	0.790862	0.812278	0.788885	0.812278	0.812154	0.812278	0.077253	0.077253	0.037685	0.997431
N2		0.003	1.38E-08	2.86E-64	6.03E-07	2.86E-64	0	2.86E-64	4.83E-07	4.83E-07	5.04E-07	9.84E-12
AR		0.001	3.65E-05	7.88E-19	4.34E-05	7.88E-19	0	7.88E-19	0.00128	0.00128	0.001335	8.73E-06
CH4		0.005	0.001163	3.39E-06	0.001209	3.39E-06	0	3.39E-06	0.040683	0.040683	0.042374	0.001361
C5H13NO2		0	0.119549	0.122774	0.119244	0.122774	0.122775	0.122774	3.72E-06	3.72E-06	2.8E-12	9.02E-05
MDEA+		0	0	0	0	0	0	0	0	0	0	0
H3O+		0	0	0	0	0	0	0	0	0	0	0
OH-		0	0	0	0	0	0	0	0	0	0	0
HCO3-		0	0	0	0	0	0	0	0	0	0	0
CO3--		0	0	0	0	0	0	0	0	0	0	0

Table 18: Stream I-11 Results for Component Mole Flows and Mole Fractions

From	Units	S12	S13	S14	S15	S16	S17	S19	S20	S21
To		ABSORBER	MIXER1	PUMP3	PUMP2	FLASH1	FLASH1	HT1	PIPE	PIPE
Substream: MIXED										
Phase:		Vapor	Liquid	Liquid	Liquid	Vapor	Liquid	Liquid	Liquid	Liquid
Component Mole Flow										
CO	KMOL/HR	539.8822	0	0.003461	1.03E-19	13.20289	28.91493	1.03E-19	0.003461	0.003461
CO2	KMOL/HR	17.85835	0	0.007264	519.5623	5.491472	697.2177	519.5623	0.007264	0.007264
H2	KMOL/HR	1194.413	0	6.38E-08	2.98E-54	1.530528	0.055986	2.98E-54	6.38E-08	6.38E-08
H2O	KMOL/HR	3.286254	9.136	18.78359	6498.264	0.339366	6497.608	6498.264	18.78359	18.7836
N2	KMOL/HR	5.995034	0	9.52E-11	2.29E-60	0.004852	0.000113	2.29E-60	9.52E-11	9.52E-11
AR	KMOL/HR	1.642756	0	8.45E-05	6.31E-15	0.005706	0.300184	6.31E-15	8.45E-05	8.45E-05
CH4	KMOL/HR	0.042115	0	0.013162	0.027108	0.399424	9.558461	0.027108	0.013162	0.013162
CSH13NO2	KMOL/HR	0.000161	0	0.000872	982.1988	2.28E-05	982.1988	982.1988	0.000872	0.000872
MDEA+	KMOL/HR	0	0	0	0	0	0	0	0	0
H3O+	KMOL/HR	0	0	0	0	0	0	0	0	0
OH-	KMOL/HR	0	0	0	0	0	0	0	0	0
HCO3-	KMOL/HR	0	0	0	0	0	0	0	0	0
CO3--	KMOL/HR	0	0	0	0	0	0	0	0	0
Component Mole Fraction										
CO		0.306208	0	0.000184	1.28E-23	0.627945	0.003519	1.28E-23	0.000184	0.000184
CO2		0.010129	0	0.000386	0.064945	0.261178	0.084862	0.064945	0.000386	0.000386
H2		0.677443	0	3.39E-09	3.73E-58	0.072794	6.81E-06	3.73E-58	3.39E-09	3.39E-09
H2O		0.001864	1	0.998679	0.812278	0.016141	0.790862	0.812278	0.998679	0.998679
N2		0.0034	0	5.06E-12	2.86E-64	0.000231	1.38E-08	2.86E-64	5.06E-12	5.06E-12
AR		0.000932	0	4.49E-06	7.88E-19	0.002714	3.65E-05	7.88E-19	4.49E-06	4.49E-06
CH4		2.39E-05	0	0.0007	3.39E-06	0.018997	0.001163	3.39E-06	0.0007	0.0007
CSH13NO2		9.13E-08	0	4.64E-05	0.122774	1.09E-06	0.119549	0.122774	4.64E-05	4.64E-05
MDEA+		0	0	0	0	0	0	0	0	0
H3O+		0	0	0	0	0	0	0	0	0
OH-		0	0	0	0	0	0	0	0	0
HCO3-		0	0	0	0	0	0	0	0	0
CO3--		0	0	0	0	0	0	0	0	0

Table 19: Stream 12-21 Results for Component Mole Flows and Mole Fractions

From	Units	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
To												
Substream: MIXED		ABSORBER	STRIPPER	EXCH	PUMP1	EXCH	ABSORBER	PUMP2	STRIPPER	HT2	FLASH2	FLASH2
Phase:		Mixed	Mixed	Liquid	Liquid	Liquid	Liquid	Liquid	Vapor	Mixed	Vapor	Liquid
Component Mass Flow												
CO	KG/HR	16302.05	809.9187	2.87E-18	1179.737	2.87E-18	0	2.87E-18	810.0156	810.0156	809.9187	0.096954
CO2	KG/HR	8801.96	30684.41	22865.84	30926.09	22865.84	22910.07	22865.83	7818.895	7818.895	7818.575	0.319694
H2	KG/HR	2410.992	0.112861	6.01E-54	3.198222	6.01E-54	0	6.01E-54	0.112861	0.112861	0.112861	1.29E-07
H2O	KG/HR	72.06112	117056	117068	117062	117068	117050	117068	326.5122	326.5122	152.7083	173.804
N2	KG/HR	168.0809	0.003176	6.4E-59	0.13911	6.4E-59	0	6.4E-59	0.003176	0.003176	0.003176	2.67E-09
AR	KG/HR	79.896	11.99176	2.52E-13	14.2712	2.52E-13	0	2.52E-13	11.99513	11.99513	11.99176	0.003375
CH4	KG/HR	160.4276	153.3441	0.434882	159.752	0.434882	0	0.43488	153.1204	153.1204	152.9092	0.211162
CSH13NO2	KG/HR	0	117043	117043	117043	117043	117043	117043	0.103963	0.103963	7.51E-08	0.103963
MDEA+	KG/HR	0	0	0	0	0	0	0	0	0	0	0
H3O+	KG/HR	0	0	0	0	0	0	0	0	0	0	0
OH-	KG/HR	0	0	0	0	0	0	0	0	0	0	0
HCO3-	KG/HR	0	0	0	0	0	0	0	0	0	0	0
CO3--	KG/HR	0	0	0	0	0	0	0	0	0	0	0
Component Mass Fraction												
CO		0.58231	0.003048	1.12E-23	0.004429	1.12E-23	0	1.12E-23	0.08881	0.08881	0.090532	0.000555
CO2		0.314407	0.11546	0.08898	0.116094	0.08898	0.089144	0.08898	0.857264	0.857264	0.873953	0.001832
H2		0.086121	4.25E-07	2.34E-59	1.2E-05	2.34E-59	0	2.34E-59	1.24E-05	1.24E-05	1.26E-05	7.37E-10
H2O		0.002574	0.440461	0.455559	0.439443	0.455559	0.455442	0.455559	0.035799	0.035799	0.01707	0.995788
N2		0.006004	1.2E-08	2.49E-64	5.22E-07	2.49E-64	0	2.49E-64	3.48E-07	3.48E-07	3.55E-07	1.53E-11
AR		0.002854	4.51E-05	9.8E-19	5.36E-05	9.8E-19	0	9.8E-19	0.001315	0.001315	0.00134	1.93E-05
CH4		0.00573	0.000577	1.69E-06	0.0006	1.69E-06	0	1.69E-06	0.016788	0.016788	0.017092	0.00121
CSH13NO2		0	0.440409	0.455459	0.439369	0.455459	0.455415	0.455459	1.14E-05	1.14E-05	8.4E-12	0.000596
MDEA+		0	0	0	0	0	0	0	0	0	0	0
H3O+		0	0	0	0	0	0	0	0	0	0	0
OH-		0	0	0	0	0	0	0	0	0	0	0
HCO3-		0	0	0	0	0	0	0	0	0	0	0
CO3--		0	0	0	0	0	0	0	0	0	0	0

Table 20: Stream 1-11 Results for Component Mass Flows and Mass Fractions

From	Units	S12	S13	S14	S15	S16	S17	S19	S20	S21
To		ABSORBER	MIXER1	PUMP3	PUMP2	FLASH1	FLASH1	HT1	PUMP3	PIPE
Substream: MIXED										
Phase:		Vapor	Liquid	Liquid	Liquid	Vapor	Liquid	Liquid	Liquid	Liquid
Component Mass Flow										
CO	KG/HR	15122.32	0	0.096954	2.87E-18	369.8182	809.9187	2.87E-18	0.096954	0.096954
CO2	KG/HR	785.9425	0	0.319694	22865.83	241.6764	30684.41	22865.83	0.319694	0.319695
H2	KG/HR	2407.794	0	1.29E-07	6.01E-54	3.085362	0.112861	6.01E-54	1.29E-07	1.29E-07
H2O	KG/HR	59.20278	164.5876	338.3916	117068	6.113769	117056	117068	338.3916	338.3917
N2	KG/HR	167.9418	0	2.67E-09	6.4E-59	0.135934	0.003176	6.4E-59	2.67E-09	2.67E-09
AR	KG/HR	65.6248	0	0.003375	2.52E-13	2.279445	11.99176	2.52E-13	0.003375	0.003375
CH4	KG/HR	0.67564	0	0.211162	0.43488	6.407868	153.3441	0.43488	0.211162	0.211162
C5H13NO2	KG/HR	0.01918	0	0.103963	117043	0.002719	117043	117043	0.103963	0.103964
MDEA+	KG/HR	0	0	0	0	0	0	0	0	0
H3O+	KG/HR	0	0	0	0	0	0	0	0	0
OH-	KG/HR	0	0	0	0	0	0	0	0	0
HCO3-	KG/HR	0	0	0	0	0	0	0	0	0
CO3--	KG/HR	0	0	0	0	0	0	0	0	0
Component Mass Fraction										
CO		0.812612	0	0.000286	1.12E-23	0.587461	0.003048	1.12E-23	0.000286	0.000286
CO2		0.042233	0	0.000943	0.08898	0.383906	0.11546	0.08898	0.000943	0.000943
H2		0.129385	0	3.79E-10	2.34E-59	0.004901	4.25E-07	2.34E-59	3.79E-10	3.79E-10
H2O		0.003181	1	0.997832	0.455559	0.009712	0.440461	0.455559	0.997832	0.997832
N2		0.009025	0	7.87E-12	2.49E-64	0.000216	1.2E-08	2.49E-64	7.87E-12	7.87E-12
AR		0.003526	0	9.95E-06	9.8E-19	0.003621	4.51E-05	9.8E-19	9.95E-06	9.95E-06
CH4		3.63E-05	0	0.000623	1.69E-06	0.010179	0.000577	1.69E-06	0.000623	0.000623
C5H13NO2		1.03E-06	0	0.000307	0.455459	4.37E-06	0.440409	0.455459	0.000307	0.000307
MDEA+		0	0	0	0	0	0	0	0	0
H3O+		0	0	0	0	0	0	0	0	0
OH-		0	0	0	0	0	0	0	0	0
HCO3-		0	0	0	0	0	0	0	0	0
CO3--		0	0	0	0	0	0	0	0	0

Table 21: Stream 12-21 Results for Component Mass Flows and Mass Fractions

	Units	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
From												
To		ABSORBER	STRIPPER	EXCH	PUMP1	ABSORBER	STRIPPER	ABSORBER	EXCH	STRIPPER	HT2	FLASH2
Substream: MIXED						FLASH1	PUMP1		PUMP2	HT2	FLASH2	MIXER1
Phase:		Mixed	Mixed	Liquid	Liquid	Liquid	Liquid	Liquid	Liquid	Vapor	Mixed	Vapor
Mole Flow	KMOL/HR	2000	8215.854	8000.056	8236.88	8000.056	8000.056	8000	8000.052	234.6069	234.6069	224.9345
Mass Flow	KG/HR	27995.47	265759	256977	266388	256977	256977	257002	256977	9120.758	9120.758	8946.219
Volume Flow	CUM/SEC	0.266813	0.067497	0.0644	0.063747	0.064394	0.064394	0.061841	0.064103	0.890252	0.807448	0.807399
Temperature	C	40	71.01537	85.34073	66.79913	85.09446	85.09446	45	81.00945	57.66883	40	40
Pressure	BAR	55	15	15	53	7.1	7.1	53	15	2	2	2
Vapor Fraction		0.999686	0.000882	0	0	0	0	0	0	1	0.958772	1
Liquid Fraction		0.000314	0.999118	1	1	1	1	1	1	0	0.041228	0
Solid Fraction		0	0	0	0	0	0	0	0	0	0	0
Molar Enthalpy	J/KMOL	-7.2E+07	-3.2E+08	-3.1E+08	-3.2E+08	-3.1E+08	-3.1E+08	-3.2E+08	-3.1E+08	-3.3E+08	-3.3E+08	-2.8E+08
Mass Enthalpy	J/KG	-5150100	9814200	9776900	-9812800	9777600	-9812800	-9869000	9785100	8551000	8613600	8473900
Enthalpy Flow	WATT	-4E+07	-7.2E+08	-7E+08	-7.3E+08	-7E+08	-7E+08	-7E+08	-7E+08	-2.2E+07	-2.2E+07	-2.1E+07
Molar Entropy	J/KMOL-K	1685.63	-310940	-291640	-312260	-291710	-307440	-307440	-293030	11316.62	3741.412	10752.61
Mass Entropy	J/KG-K	120.4216	-9612.67	-9079.11	-9655.23	-9081.45	-9570.12	-9570.12	-9122.39	291.0896	96.23775	270.3524
Molar Density	KMOL/CUM	2.082188	33.81143	34.50701	35.89233	34.51025	35.93429	34.66665	34.66665	0.073202	0.080709	0.077387
Mass Density	KG/CUM	29.14592	1093.7	1108.43	1160.79	1108.534	1154.398	1113.559	1113.559	2.845874	3.137717	3.077861
Average Molecular Weight		13.99774	32.34703	32.12189	32.34089	32.12189	32.12526	32.12526	32.12189	38.87677	38.87677	39.77256
Phase: All												
Mole flow rate, mixture	MMSCFD	40.15758	164.9644	160.6314	165.3866	160.6314	160.6314	160.6303	160.6314	4.710623	4.710623	4.516412
												0.194211

Table 22: Stream 1-11 Miscellaneous Results

	Units	S12	S13	S14	S15	S16	S17	S19	S20	S21
From		ABSORBER		MIXER1	PUMP2	FLASH1	FLASH1	HT1	PUMP3	PIPE
To			MIXER1	PUMP3	HT1		EXCH		PIPE	STRIPPER
Substream: MIXED										
Phase:		Vapor	Liquid	Liquid	Liquid	Vapor	Liquid	Liquid	Liquid	Liquid
Mole Flow	KMOL/HR	1763.12	9.136	18.80843	8000.052	21.02557	8215.854	8000.052	18.80843	18.80844
Mass Flow	KG/HR	18609.52	164.5876	339.1267	256977	629.5197	265759	256977	339.1267	339.1269
Volume Flow	CUM/SEC	0.250384	4.61E-05	9.54E-05	0.064128	0.010861	0.063497	0.06185	9.54E-05	9.54E-05
Temperature	C	45.0528	39.95	39.97562	82.21275	66.31932	66.31932	45	39.97934	39.97934
Pressure	BAR	53	2	2	55	15	15	53	3.2	2.077977
Vapor Fraction		1	0	0	0	1	0	0	0	0
Liquid Fraction		0	1	1	1	0	1	1	1	1
Solid Fraction		0	0	0	0	0	0	0	0	0
Molar Enthalpy	J/KMOL	-3.8E+07	-2.8E+08	-2.8E+08	-3.1E+08	-1.8E+08	-3.2E+08	-3.2E+08	-2.8E+08	-2.8E+08
Mass Enthalpy	J/KG	-3576600	-1.6E+07	-1.6E+07	9781500	5894000	9822100	9869300	-1.6E+07	-1.6E+07
Enthalpy Flow	WATT	-1.8E+07	-722960	1487700	-7E+08	1030700	-7.3E+08	-7E+08	1487700	1487700
Molar Entropy	J/KMOL-K	2063.485	-159480	-159380	-292660	44498.41	-312680	-307320	-159380	-159380
Mass Entropy	J/KG-K	195.5006	-8852.62	-8839.66	-9110.94	1486.219	-9666.32	-9567.44	-8839.7	-8839.61
Molar Density	KMOL/CUM	1.956016	55.0844	54.78497	34.65298	0.537734	35.94135	35.92952	54.78779	54.78508
Mass Density	KG/CUM	20.6455	992.3609	987.8043	1113.119	16.10011	1162.596	1154.124	987.855	987.8062
Average Molecular Weight		10.55488	18.01528	18.03057	32.12189	29.94068	32.34703	32.12189	18.03057	18.03057
Phase: All										
Mole flow rate, mixture	MMSCFD	35.40132	0.18344	0.377651	160.6314	0.422168	164.9644	160.6314	0.377651	0.377651

Table 23: Stream 12-21 Miscellaneous Results

IX OPERATING COSTS

	Electricity	Cooling	Heating
Cost	0.06 \$/kWh	0.2 \$/GJ	5.5 \$/GJ
PUMP1	50.8709068 kW		
PUMP2	256.411991 kW		
PUMP3	0.011443802 kW		
HT1		-22.56386702 GJ/hr	
HT2		-0.570777113 GJ/hr	
S Reboiler			22.94746481 GJ/hr
Operating Hours	8000 hr/yr	8000 hr/yr	8000 hr/yr
Annual Cost	\$ 147,501.28 \$/yr	\$ 37,015.43 \$/yr	\$ 1,009,688.45 \$/yr

Total Annual Cost [\$ /yr] = \$ 1,194,205.17

Table 24: Operating Cost Calculations from Excel

Calculations:

$$\text{Annual Utility Cost} \left[\frac{\$}{\text{Year}} \right] = \text{Cost} \left[\frac{\$}{\text{GJ}} \right] * \text{Total } Q \left[\frac{\text{GJ}}{\text{Hour}} \right] * \frac{\text{Operating Hours}}{\text{Year}}$$

$$\text{Cold Utilities: } (-1) * 0.2 \left(\frac{\$}{\text{GJ}} \right) * (-23.1) \left(\frac{\text{GJ}}{\text{Hour}} \right) * \frac{8000 \text{ Hours}}{\text{Year}} = \frac{\$37,015}{\text{Year}}$$

$$\text{Hot Utilities: } 5.5 \left(\frac{\$}{\text{GJ}} \right) * (22.95) \left(\frac{\text{GJ}}{\text{Hour}} \right) * \frac{8000 \text{ Hours}}{\text{Year}} = \frac{\$1,009,688}{\text{Year}}$$

$$\text{Annual Electricity Cost} \left[\frac{\$}{\text{Year}} \right] = \text{Cost} \left[\frac{\$}{\text{kWh}} \right] * \text{Total } W [\text{kW}] * \frac{\text{Operating Hours}}{\text{Year}}$$

$$\text{Electricity: } 0.06 \left(\frac{\$}{\text{kWh}} \right) * 307.3 \text{ kW} * \frac{8000 \text{ Hours}}{\text{Year}} = \frac{\$147,501}{\text{Year}}$$

$$\text{Total Annual Cost} \left[\frac{\$}{\text{Year}} \right] = (37,015 + 1,009,688 + 147,501) \frac{\$}{\text{Year}} = \frac{\$1,194,205}{\text{Year}}$$

X STRIPPER COLUMN SIZING

Using the Equipment Sizing feature on ASPEN Plus, the height of the column was calculated as shown below (62 feet, which is about 18.9 m).

Name	DTW TRAYED STRIPP
User tag number	STRIPPER-tower
Remarks 1	Equipment mapped
Quoted cost per item [USD]	
Currency unit for matl cost	▼
Number of identical items	
Installation option	▼
Tray type	SIEVE ▼
Application	▼
Shell material	▼
Vessel diameter [FEET]	2
Vessel tangent to tangent height [FEET]	62
Design gauge pressure [PSIG]	113.277
Vacuum design gauge pressure [PSIG]	
Design temperature [DEG F]	250
Operating temperature [DEG F]	185.17

Figure 39: Stripper Column Sizing
